

Daffodil International University

DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING

Dhaka, BANGLADESH.

COURSE NO.: ICE 3242

COURSE TITLE: DIGITAL SIGNAL PROCESSING Lab

Experiment No. 03

Name of Experiment: Addition, Multiplication, Energy and Power of Discrete-Time Signals

Objectives:

The objectives of this experiment are:

- To add and multiply two signals that start and end at different time indices.
- To find the energy of a short (finite) signal and of a signal that goes on forever but dies out.
- To find the average power of a sinusoid and of the unit step.
- To understand the difference between the power of a few samples and the true average power of a signal.
- To decide whether a signal is an energy signal, a power signal or neither, and to see how scaling, shifting and reversing affect energy.

THEORY:

In Experiment 01 we made the basic signals $\delta(n)$, $u(n)$, $r(n)$ and $a^n u(n)$, and in Experiment 02 we learned that a signal in MATLAB needs two vectors: the time index n and the values x . In this experiment we first *add* and *multiply* two signals, and then measure how *big* a signal is by one number — its **energy** or its **power**.

1. Signal Addition

To add two signals, we add the values that are at the **same time index** n :

$$y(n) = x_1(n) + x_2(n)$$

MATLAB adds two vectors by **position** (1st with 1st, 2nd with 2nd ...), not by time index. So $x_1 + x_2$ is correct only if both signals start at the same n and have the same length. If not, we must line them up first:

1. Find the common range: from the smallest start index to the largest end index.
2. Add zeros before and/or after each signal so that both cover the common range. This is called **zero-padding** (a signal is zero outside its own range).
3. Now both vectors have the same length and each position means the same n .
4. Add (or multiply) the two padded vectors element by element.

In this experiment $x_1(n) = \{1, 2, 3, 4, 5\}$ for $-2 \leq n \leq 2$ and $x_2(n) = \{3, -1, 2, 1\}$ for $0 \leq n \leq 3$. The common range is $-2 \leq n \leq 3$. So x_1 needs **one zero at the end** ($n = 3$) and x_2 needs

two zeros at the front ($n = -2, -1$). Table 3.1 shows the padded signals and the results, and Fig. 3.1 shows the output of Program 1.

Table 3.1 Alignment of $x_1(n)$ and $x_2(n)$ on the common range $-2 \leq n \leq 3$

n	-2	-1	0	1	2	3
$x_1(n)$ (padded)	1	2	3	4	5	0
$x_2(n)$ (padded)	0	0	3	-1	2	1
$x_1(n) + x_2(n)$	1	2	6	3	7	1
$x_1(n) \cdot x_2(n)$	0	0	9	-4	10	0

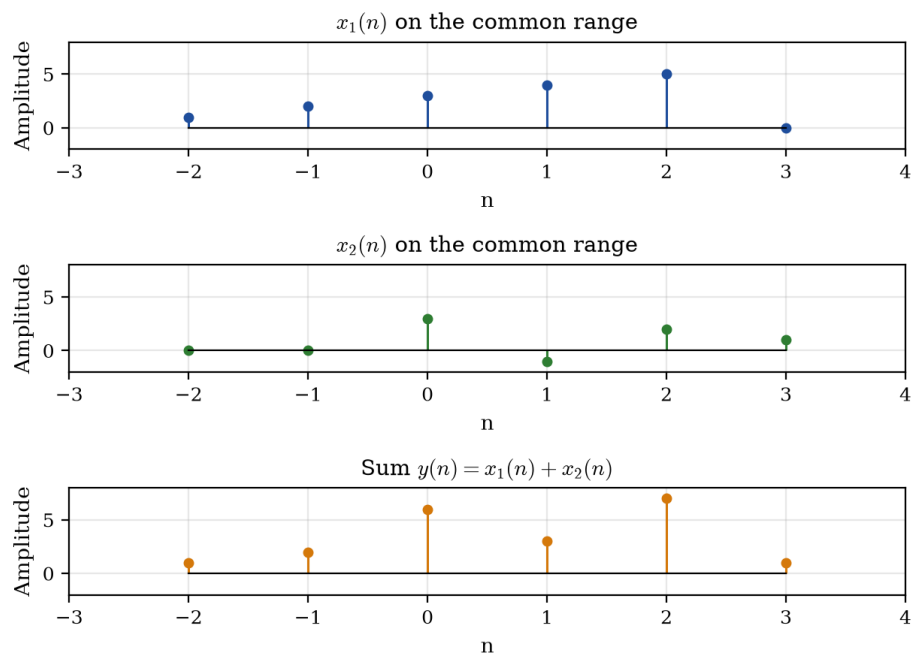


Fig. 3.1 Output of Program 1 – zero-padded $x_1(n)$, $x_2(n)$ and their sum on $-2 \leq n \leq 3$

Read Table 3.1 column by column: at $n = -2$ and -1 only x_1 has a value, at $n = 3$ only x_2 has a value, and for $0 \leq n \leq 2$ both values are added. Result: $y(n) = \{1, 2, 6, 3, 7, 1\}$.

2. Signal Multiplication

Multiplication is also done sample by sample:

$$y(n) = x_1(n) \cdot x_2(n)$$

Multiplication is used in *modulation* (message $\times$ carrier) and *windowing* (keeping only one part of a signal). The signals are zero-padded in the same way, and in MATLAB we use `.*` (element-by-element multiply), not `*`. Since zero times anything is zero, the product is non-zero only where **both** signals are non-zero ($0 \leq n \leq 2$). Result: $y(n) = \{0, 0, 9, -4, 10, 0\}$ (Fig. 3.2).

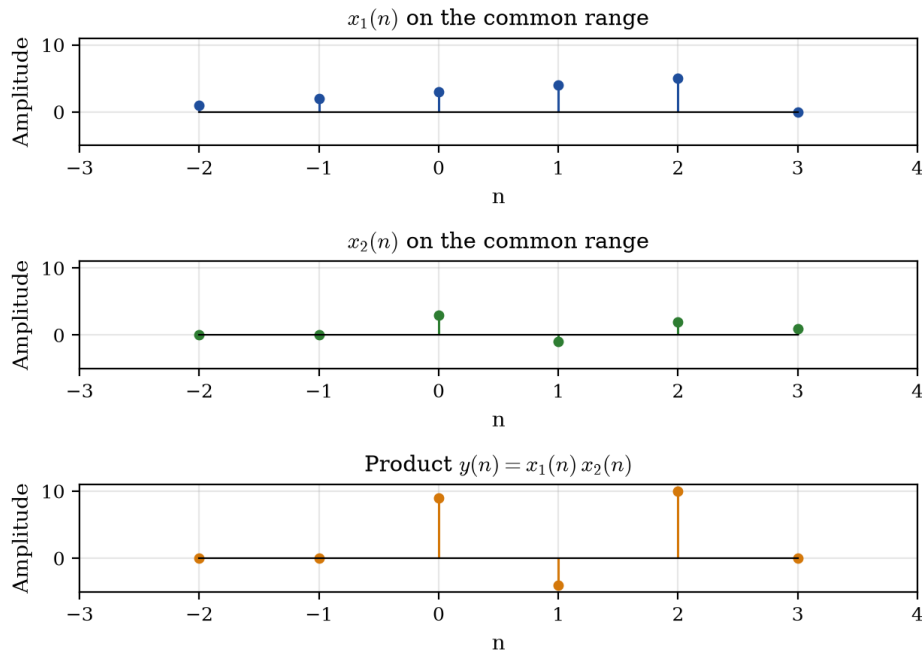


Fig. 3.2 Output of Program 2 – zero-padded $x_1(n)$, $x_2(n)$ and their product

3. Energy of a Discrete-Time Signal

The energy of a signal is the sum of the squares of all its samples:

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

($|x(n)|$ means magnitude; it only matters for complex signals.) If the signal is non-zero only from $n = N_1$ to N_2 , we just add the squares over that range:

$$E = \sum_{n=N_1}^{N_2} |x(n)|^2$$

Example: $E_1 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$ and $E_2 = 3^2 + (-1)^2 + 2^2 + 1^2 = 15$. In MATLAB: `E = sum(x.^2)`. For a complex signal use `sum(abs(x).^2)`.

Energy depends only on the sample **values**, not on where they sit on the n -axis:

- **Scaling:** multiplying a signal by A multiplies its energy by A^2 . For $A = 3$: $E = 9 \times 55 = 495$.
- **Shifting and reversing:** $x(n - k)$ and $x(-n)$ have the same values as $x(n)$, only at other positions, so the energy does not change (55).

- **Adding:** energies do **not** simply add. $E\{x_1 + x_2\} = 100$, not $55 + 15 = 70$, because of an extra cross term $2 \sum x_1(n)x_2(n) = 2 \times 15 = 30$. The cross term is zero only when the two signals do not overlap (more exactly, when they are orthogonal).

Fig. 3.3 shows the output of Program 3. The energy of each signal is written in its title.

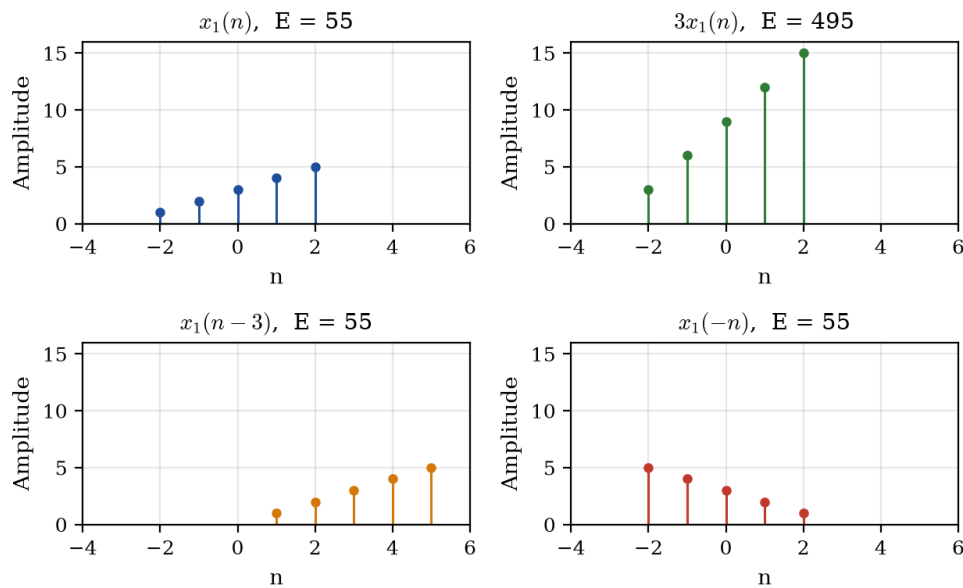


Fig. 3.3 Output of Program 3 – scaling multiplies the energy by $A^2 = 9$; shifting and reversal leave it unchanged

4. Energy of an Infinite-Length Sequence

Some signals go on forever, for example $x(n) = a^n u(n)$. If $|a| < 1$ the samples get smaller and smaller, so the energy is a finite number (a geometric series):

$$E = \sum_{n=0}^{\infty} |a|^{2n} = \frac{1}{1 - |a|^2}, \quad |a| < 1$$

A computer cannot add infinitely many terms, so we add only the first $N + 1$ terms (a *partial sum*):

$$E_N = \sum_{n=0}^N |a|^{2n} = \frac{1 - |a|^{2(N+1)}}{1 - |a|^2}$$

As N grows, E_N comes closer to the exact value. For $a = 0.8$ the exact energy is $1/(1 - 0.64) = 2.7778$, and the partial sum is 2.5869 for $N = 5$, 2.7573 for $N = 10$ and 2.777775 for $N = 30$. In MATLAB, `cumsum` gives all the partial sums in one step. Fig. 3.4 shows the output of Program 4.

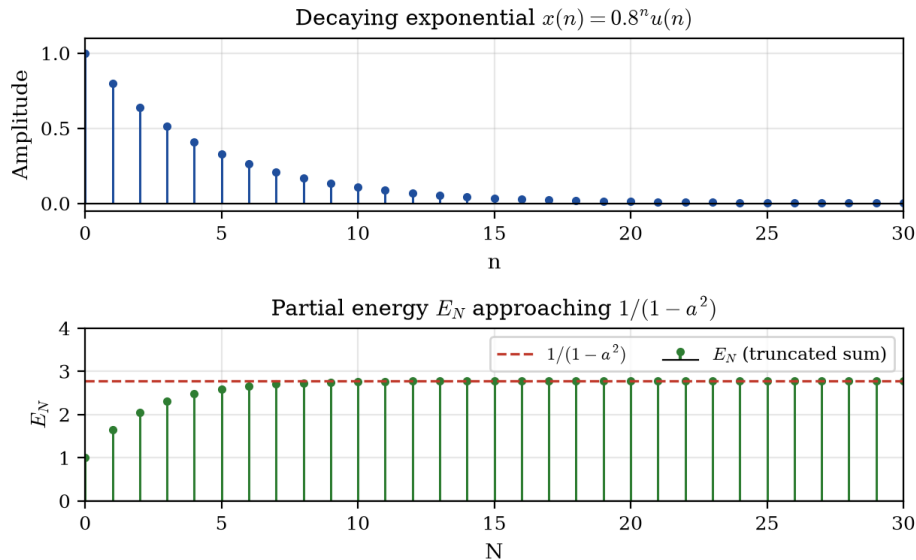


Fig. 3.4 Output of Program 4 – $0.8^n u(n)$ and its partial energy E_N converging to $1/(1 - 0.64) = 2.7778$

5. Average Power and the Power of a Sinusoid

Signals such as a sinusoid or a unit step never die out, so their energy is infinite. For them we use the **average power**, i.e. the *energy per sample*, taken over a window $-N \leq n \leq N$ that becomes very large:

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

For a **periodic** signal with period N , every period is the same. So we only need the average over one period (any whole number of periods gives the same answer):

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

For a **sinusoid** $x(n) = A \cos(\omega_0 n + \phi)$ the power is simply $P = A^2/2$ (special case: $\cos(\pi n) = (-1)^n$ has power 1, because its square is always 1). Program 5 uses $x(n) = 2 \cos(2\pi n/16)$, so $P = 4/2 = 2$. It also finds the *running power*, i.e. the average of $x^2(n)$ over the first L samples. Fig. 3.5 shows that it starts at 4, swings around 2 and settles at 2. It equals 2 exactly whenever L is a multiple of 8. So power must be measured over whole periods or over many samples.

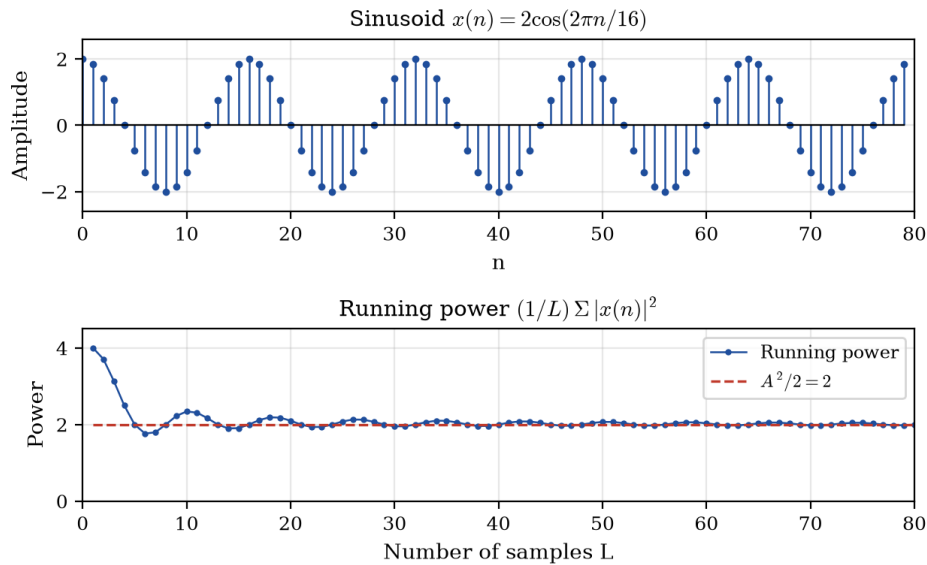


Fig. 3.5 Output of Program 5 – sinusoid of amplitude 2 and its running power settling to $A^2/2 = 2$

6. Power of the Unit Step

The unit step is not periodic, so we use the definition directly. In the window $-N \leq n \leq N$ there are $N + 1$ ones ($n = 0 \dots N$) and N zeros ($n < 0$), so

$$P_N = \frac{N+1}{2N+1}, \quad P = \lim_{N \rightarrow \infty} \frac{N+1}{2N+1} = \frac{1}{2}$$

For $N = 10, 100$ and 1000 this gives $0.52381, 0.50249$ and 0.50025 , which gets closer and closer to $1/2$. The answer is $1/2$ because half of the window ($n < 0$) has $u(n) = 0$. So the unit step has infinite energy and power $P = 1/2$. Fig. 3.6 shows P_N from Program 6.

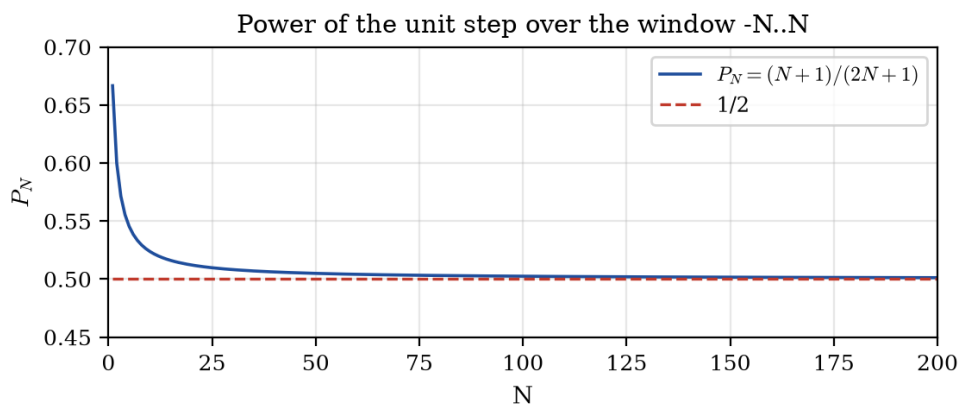


Fig. 3.6 Output of Program 6 – power of $u(n)$ over $-N \leq n \leq N$ approaching $1/2$

Note: Some textbooks average only over $n = 0 \dots N - 1$. With that definition the unit step has $P = 1$ instead of $1/2$. For periodic signals both definitions give the same answer.

7. Power of a Finite Block of Samples

In practice we only record a finite block of L samples. The ‘power’ of this block is its energy divided by L (the mean-square value):

$$P_L = \frac{1}{L} \sum_{n=n_0}^{n_0+L-1} |x(n)|^2 = \frac{E_{\text{block}}}{L}$$

For $x_1(n)$: $L = 5$ and $E_1 = 55$, so $P_L = 11$ (Program 3). But this is only the power **of these 5 samples**. If $x_1(n)$ is zero everywhere else, its true average power is $55/(2N + 1)$, which goes to **0** as $N \rightarrow \infty$. Block power is useful when the block is a piece of a long signal, for example a recorded sinusoid or noise.

8. Energy Signals, Power Signals and Neither

Using E and P , every signal belongs to one of three classes:

Table 3.2 Classification of discrete-time signals by energy and power

Class	Condition	Examples
Energy signal	$0 < E < \infty$, hence $P = 0$	any finite-length sequence; $a^n u(n)$ with $ a < 1$
Power signal	$0 < P < \infty$, hence $E = \infty$	sinusoids; periodic sequences; $u(n)$ ($P = 1/2$)
Neither	$E = \infty$ and $P = \infty$	ramp $r(n) = n u(n)$; $a^n u(n)$ with $ a > 1$

A non-zero signal cannot be both: if E is finite, then $P = E/(2N + 1) \rightarrow 0$; if P is greater than 0, the energy keeps growing and becomes infinite. For the ramp both E and P grow without limit. Fig. 3.7 shows one example of each class.

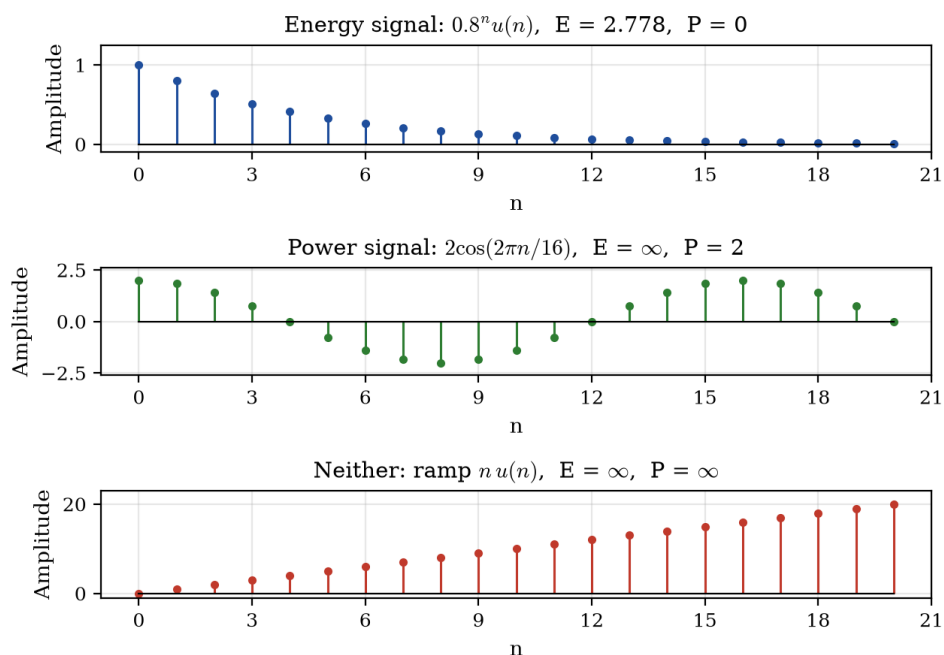


Fig. 3.7 Typical energy signal, power signal and a signal that is neither

Important Equations

$$y(n) = x_1(n) + x_2(n), \quad n_{\min} \leq n \leq n_{\max} \quad (\text{addition}) \quad (3.1)$$

$$y(n) = x_1(n) \cdot x_2(n) \quad (\text{multiplication}) \quad (3.2)$$

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \quad (\text{energy}) \quad (3.3)$$

$$E = \sum_{n=N_1}^{N_2} |x(n)|^2 \quad (\text{finite sequence}) \quad (3.4)$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 \quad (\text{average power}) \quad (3.5)$$

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \quad (\text{periodic, period } N) \quad (3.6)$$

$$P_L = \frac{1}{L} \sum_{n=n_0}^{n_0+L-1} |x(n)|^2 \quad (\text{power over } L \text{ samples}) \quad (3.7)$$

$$\sum_{n=0}^{\infty} |a^n|^2 = \frac{1}{1-|a|^2}, \quad |a| < 1 \quad (\text{energy of } a^n u(n)) \quad (3.8)$$

$$P = \frac{A^2}{2} \quad \text{for } x(n) = A \cos(\omega_0 n + \phi), \quad 0 < \omega_0 < \pi \quad (3.9)$$

$$P = \lim_{N \rightarrow \infty} \frac{N+1}{2N+1} = \frac{1}{2} \quad (\text{unit step}) \quad (3.10)$$

$$E\{A x(n)\} = |A|^2 E, \quad E\{x(n-k)\} = E\{x(-n)\} = E \quad (3.11)$$

$$E\{x_1 + x_2\} = E_1 + E_2 + 2 \sum_n x_1(n) x_2(n) \quad (\text{real signals}) \quad (3.12)$$

MATLAB commands used

- `[x1 0], [0 0 x2]` – joins vectors; used here to add zeros (zero-padding).
- `.*` `.^` `./` – element-by-element multiply, power and divide (sample-by-sample).
- `sum(x)` – adds all elements; `sum(x.^2)` gives the energy.
- `cumsum(x)` – running sum; gives all partial sums at once.
- `length(x)` – number of samples in a vector.
- `fliplr(x)` – reverses a vector (time reversal).
- `zeros(1,N)` – row vector of N zeros; reserves space for results.
- `double(n >= 0)` – changes true/false into 1/0; gives the unit step.
- `stem(n, x)` – plots a discrete-time signal.
- `subplot(r,c,k)` – puts several plots in one figure window.

- `num2str(v)` – changes a number into text (for plot titles).
- `fprintf('E = %g\n', E)` – prints a value in the command window.
- `hold on` / `hold off` – draws another line on the same plot.
- `legend('a', 'b')` – labels the lines in a plot.

PROCEDURE:

Apparatus Required

- A computer with MATLAB 7.0 or later (no toolbox is needed).

Algorithm

1. Open MATLAB and create a new script (M-file) for each program. Each program starts with `clc; clear all; close all;`.
2. **Program 1 (addition):** enter $x_1(n)$ and $x_2(n)$ with their index vectors, add zeros so that both cover $n = -2 \dots 3$, add them and plot.
3. **Program 2 (multiplication):** zero-pad the signals in the same way, multiply them with `.*` and plot.
4. **Program 3 (energy):** find E_1 , E_2 , the energy of $x_1 + x_2$, the power of the 5 samples of x_1 , and the energy of $3x_1(n)$, $x_1(n - 3)$ and $x_1(-n)$.
5. **Program 4 (energy of $0.8^n u(n)$):** make $x(n) = 0.8^n$ for $n = 0 \dots 30$, find the partial energies with `cumsum` and compare with $1/(1 - a^2)$.
6. **Program 5 (power of a sinusoid):** make $x(n) = 2 \cos(2\pi n/16)$ for 5 whole periods, find the power over one period and over all periods, and plot the running power.
7. **Program 6 (power of the unit step):** for $N = 1 \dots 1000$ find $P_N = (\text{number of ones})/(2N + 1)$, plot it and print it for $N = 10, 100$ and 1000 .
8. Write the displayed values in the observation table and compare them with the expected values.

Program

1. Signal Addition

```
% 1. SIGNAL ADDITION
clc; clear all; close all;

n1 = -2:2;    x1 = [1 2 3 4 5];    % x1(n), -2 <= n <= 2
n2 = 0:3;     x2 = [3 -1 2 1];    % x2(n), 0 <= n <= 3

n = -2:3;     % common range: -2 to 3

% Zero-padding: both signals now cover n = -2 ... 3
y1 = [x1 0];  % one zero at the end (n = 3)
y2 = [0 0 x2]; % two zeros at the front (n = -2, -1)

y = y1 + y2;  % y(n) = x1(n) + x2(n)

subplot(3,1,1); stem(n, y1); grid on; axis([-3 4 -2 8]);
title('x_1(n) on the common range'); xlabel('n'); ylabel('Amplitude');

subplot(3,1,2); stem(n, y2); grid on; axis([-3 4 -2 8]);
```

```

title('x_2(n) on the common range'); xlabel('n'); ylabel('Amplitude');

subplot(3,1,3); stem(n, y); grid on; axis([-3 4 -2 8]);
title('Sum y(n) = x_1(n) + x_2(n)'); xlabel('n'); ylabel('Amplitude');

disp('Row 1 = n,   Row 2 = y(n)');
disp([n; y]);

```

2. Signal Multiplication

```

% 2. SIGNAL MULTIPLICATION
clc; clear all; close all;

n1 = -2:2;   x1 = [1 2 3 4 5];   % x1(n), -2 <= n <= 2
n2 = 0:3;    x2 = [3 -1 2 1];    % x2(n), 0 <= n <= 3

n = -2:3;    % common range
y1 = [x1 0]; % zero-padded x1
y2 = [0 0 x2]; % zero-padded x2

y = y1 .* y2; % y(n) = x1(n) x2(n) (note the dot)

subplot(3,1,1); stem(n, y1); grid on; axis([-3 4 -5 11]);
title('x_1(n) on the common range'); xlabel('n'); ylabel('Amplitude');

subplot(3,1,2); stem(n, y2); grid on; axis([-3 4 -5 11]);
title('x_2(n) on the common range'); xlabel('n'); ylabel('Amplitude');

subplot(3,1,3); stem(n, y); grid on; axis([-3 4 -5 11]);
title('Product y(n) = x_1(n) x_2(n)'); xlabel('n'); ylabel('Amplitude');

disp('Row 1 = n,   Row 2 = y(n)');
disp([n; y]);

```

3. Energy of Finite Sequences

```

% 3. ENERGY OF FINITE SEQUENCES
clc; clear all; close all;

n1 = -2:2;   x1 = [1 2 3 4 5];   % x1(n), -2 <= n <= 2
n2 = 0:3;    x2 = [3 -1 2 1];    % x2(n), 0 <= n <= 3

E1 = sum(x1.^2); % energy of x1 = 55
E2 = sum(x2.^2); % energy of x2 = 15

y = [x1 0] + [0 0 x2]; % x1(n) + x2(n) (as in Program 1)
E_add = sum(y.^2); % energy of the sum = 100

L1 = length(x1); % number of samples = 5
P1 = E1/L1; % power of these 5 samples = 11

xa = 3*x1; % scaling: 3x1(n)
Ea = sum(xa.^2); % = 9 x 55 = 495

n_sh = n1 + 3; % shifting: x1(n-3), same values
E_sh = sum(x1.^2); % = 55

n_rev = -fliplr(n1); % reversing: x1(-n)
x_rev = fliplr(x1);
E_rev = sum(x_rev.^2); % = 55

```

```

subplot(2,2,1); stem(n1, x1); grid on; axis([-4 6 0 16]);
title(['x_1(n), E = ' num2str(E1)]); xlabel('n');
subplot(2,2,2); stem(n1, xa); grid on; axis([-4 6 0 16]);
title(['3x_1(n), E = ' num2str(Ea)]); xlabel('n');
subplot(2,2,3); stem(n_sh, x1); grid on; axis([-4 6 0 16]);
title(['x_1(n-3), E = ' num2str(E_sh)]); xlabel('n');
subplot(2,2,4); stem(n_rev, x_rev); grid on; axis([-4 6 0 16]);
title(['x_1(-n), E = ' num2str(E_rev)]); xlabel('n');

fprintf('E1 = %g, E2 = %g\n', E1, E2);
fprintf('E(x1+x2) = %g, E1+E2 = %g\n', E_add, E1 + E2);
fprintf('Power of x1 over %d samples = %g\n', L1, P1);
fprintf('E(3x1) = %g, 9*E1 = %g\n', Ea, 9*E1);
fprintf('E(x1(n-3)) = %g, E(x1(-n)) = %g\n', E_sh, E_rev);

```

4. Energy of a Decaying Exponential

```

% 4. ENERGY OF A DECAYING EXPONENTIAL
clc; clear all; close all;

a = 0.8;
n = 0:30; % first 31 samples
x = a.^n; % x(n) = 0.8^n u(n)

E_N = cumsum(x.^2); % partial energies E_0, E_1, ..., E_30
E_exact = 1/(1 - a^2); % exact energy = 2.7778

subplot(2,1,1); stem(n, x); grid on;
title('Decaying exponential x(n) = 0.8^n u(n)');
xlabel('n'); ylabel('Amplitude');

subplot(2,1,2); stem(n, E_N); grid on; hold on;
plot([0 30], [E_exact E_exact], 'r--'); % line at exact value
hold off; axis([0 30 0 4]);
title('Partial energy E_N approaching 1/(1-a^2)');
xlabel('N'); ylabel('E_N');
legend('E_N (truncated sum)', '1/(1-a^2)');

% MATLAB index starts at 1, so N = 30 is element 31
fprintf('E_N(N=30) = %.6f, 1/(1-a^2) = %.6f\n', E_N(31), E_exact);

```

5. Power of a Sinusoid

```

% 5. POWER OF A SINUSOID
clc; clear all; close all;

A = 2; % amplitude
N0 = 16; % period (in samples)
n = 0:79; % 5 whole periods = 80 samples
x = A*cos(2*pi*n/N0); % x(n) = 2cos(2*pi*n/16)

P_one = sum(x(1:16).^2)/16; % power over one period
P_all = sum(x.^2)/80; % power over 5 periods
P_th = A^2/2; % theory: A^2/2 = 2

L = 1:80; % number of samples used
P_run = cumsum(x.^2)./L; % running power

subplot(2,1,1); stem(n, x); grid on;
title('Sinusoid x(n) = 2cos(2\pi n/16)');
xlabel('n'); ylabel('Amplitude');

```

```

subplot(2,1,2); plot(L, P_run, 'b.-'); grid on; hold on;
plot([0 80], [P_th P_th], 'r--'); % line at A^2/2
hold off; axis([0 80 0 4.5]);
title('Running power (1/L) \Sigma |x(n)|^2');
xlabel('Number of samples L'); ylabel('Power');
legend('Running power', 'A^2/2 = 2');

fprintf('P one period = %g, P 5 periods = %g\n', P_one, P_all);
fprintf('Theoretical power A^2/2 = %g\n', P_th);

```

6. Power of the Unit Step

```

% 6. POWER OF THE UNIT STEP
clc; clear all; close all;

P = zeros(1, 1000); % space for P_N, N = 1 ... 1000

for N = 1:1000
    n = -N:N; % window -N ... N
    u = double(n >= 0); % unit step u(n)
    P(N) = sum(u.^2)/(2*N + 1); % power in this window
end

plot(1:200, P(1:200), 'b'); grid on; hold on;
plot([0 200], [0.5 0.5], 'r--'); % limit = 1/2
hold off; axis([0 200 0.45 0.7]);
title('Power of the unit step over the window -N..N');
xlabel('N'); ylabel('P_N');
legend('P_N = (N+1)/(2N+1)', '1/2');

fprintf('Unit step, N = 10 : P = %.5f\n', P(10));
fprintf('Unit step, N = 100 : P = %.5f\n', P(100));
fprintf('Unit step, N = 1000 : P = %.5f\n', P(1000));

```

Program Explanation

MATLAB statement	What it does	Why it is used
% Program 1 and 2		
clc; clear all; close all;	Clears the command window, all variables and all figures.	So that results of an earlier run do not mix with the new ones.
n1 = -2:2; x1 = [1 2 3 4 5];	Stores x1(n) as two vectors: the index n1 and the values x1.	A signal always needs both its time index and its values.
n2 = 0:3; x2 = [3 -1 2 1];	Stores x2(n), which starts at a different index and is shorter.	So x1 + x2 cannot be used directly.
n = -2:3;	Common range: from the smallest start (-2) to the largest end (3).	Both signals must use the same indices.
y1 = [x1 0];	Adds one zero after x1 (for n = 3): [1 2 3 4 5 0].	Zero-padding; x1 is zero outside its own range.
y2 = [0 0 x2];	Adds two zeros before x2 (for n = -2, -1): [0 0 3 -1 2 1].	Now y1 and y2 have 6 samples each and line up.
y = y1 + y2;	Adds sample by sample: [1 2 6 3 7 1].	Gives x1(n) + x2(n).

MATLAB statement	What it does	Why it is used
<code>y = y1 .* y2;</code>	Multiplies sample by sample: [0 0 9 -4 10 0].	The dot is needed; * means matrix multiplication and gives an error.
<code>subplot(3,1,k); stem(n, ...);</code>	Draws three plots one below the other with the same axes.	To compare the signals and the result sample by sample.
<code>disp([n; y]);</code>	Prints n in row 1 and y(n) in row 2.	To check the result with Table 3.1.
<code>% Program 3</code>		
<code>E1 = sum(x1.^2);</code>	Squares every sample and adds them: E1 = 55, E2 = 15.	This is the energy formula.
<code>y = [x1 0] + [0 0 x2]; E_add = sum(y.^2);</code>	Energy of x1 + x2 = 100.	Shows that energy does not simply add (55 + 15 = 70).
<code>L1 = length(x1); P1 = E1/L1;</code>	Energy divided by the number of samples: P1 = 11.	Power of this block of 5 samples only.
<code>xa = 3*x1; Ea = sum(xa.^2);</code>	Energy of 3x1(n) = 495.	Checks that scaling by 3 multiplies the energy by 9.
<code>n_sh = n1 + 3;</code>	Index of x1(n - 3); the values stay the same.	Shifting does not change the energy (55).
<code>n_rev = -fliplr(n1); x_rev = fliplr(x1);</code>	Makes x1(-n).	Reversing does not change the energy (55).
<code>title(['x_1(n), E = ' num2str(E1)]);</code>	Joins text and the number into one title.	Shows the energy on each plot.
<code>% Program 4</code>		
<code>x = a.^n;</code>	Makes 31 samples of 0.8 ⁿ (n = 0 ... 30).	The infinite signal is cut at n = 30.
<code>E_N = cumsum(x.^2);</code>	Running sum of x ² (n); element k is the energy of the first k samples.	Shows how the partial energy grows. E_N(31) is N = 30 because MATLAB starts counting at 1.
<code>E_exact = 1/(1 - a^2);</code>	Exact energy = 2.7778.	The value E_N should approach.
<code>hold on; plot([0 30], [E_exact E_exact], 'r--');</code>	Draws a red dashed line at the exact energy.	To see E_N come close to the exact value.
<code>% Program 5</code>		
<code>n = 0:79; x = A*cos(2*pi*n/N0);</code>	Makes 80 samples (5 whole periods) of 2 cos(2πn/16).	A signal whose power is known: A ² /2 = 2.
<code>P_one = sum(x(1:16).^2)/16;</code>	Average of x ² (n) over one period.	Power of a periodic signal.
<code>P_all = sum(x.^2)/80;</code>	Average over all 5 periods.	Any whole number of periods gives the same power.
<code>P_run = cumsum(x.^2)./L;</code>	Average of x ² (n) over the first L samples, L = 1 ... 80.	Shows the power settling at 2 as more samples are used.

MATLAB statement	What it does	Why it is used
% Program 6		
for N = 1:1000 ... end	Repeats the calculation for N = 1, 2, ..., 1000.	To see what happens when the window gets bigger.
u = double(n >= 0);	Gives 1 for $n \geq 0$ and 0 for $n < 0$.	Makes the unit step in the window.
P(N) = sum(u.^2)/(2*N + 1);	Number of ones (N + 1) divided by the window length (2N + 1).	Gives $(N + 1)/(2N + 1)$, which goes to 1/2.
fprintf('... %.5f\n', P(10));	Prints the value with 5 decimal places.	Values for the observation table.

Precautions

- **Line up before adding:** zero-pad both signals to the same index range first. Equal length alone is not enough.
- **Use the dot:** write $x1.*x2$, $x.^2$ and $a./b$ for sample-by-sample operations.
- **Complex signals:** use $\text{sum}(\text{abs}(x).^2)$ for the energy, not $\text{sum}(x.^2)$.
- **Whole periods:** find the power of a periodic signal over whole periods (or over many samples).
- **Block power is not average power:** E/L of a short signal is not its true average power (which is 0).
- **Energy does not add:** $E\{x1 + x2\}$ is usually not equal to $E1 + E2$.
- **Enough terms:** for an infinite signal, keep adding terms until E_N stops changing.
- **MATLAB index starts at 1:** in Program 4, $E_N(1)$ is $N = 0$, so $N = 30$ is $E_N(31)$.

RESULTS:

Run each program in MATLAB and paste the resulting figure in the space provided below.

S.No.	Program	Output Waveform (paste figure here)
1	Signal Addition	
2	Signal Multiplication	

S.No.	Program	Output Waveform (paste figure here)
3	Energy of Finite Sequences	
4	Energy of a Decaying Exponential	
5	Power of a Sinusoid	
6	Power of the Unit Step	

Expected Output

- **Program 1:** both padded signals are drawn on $-2 \leq n \leq 3$; the sum is $y(n) = \{1, 2, 6, 3, 7, 1\}$.
- **Program 2:** the product is $y(n) = \{0, 0, 9, -4, 10, 0\}$; it is non-zero only for $0 \leq n \leq 2$.
- **Program 3:** $E_1 = 55$, $E_2 = 15$, $E\{x_1 + x_2\} = 100$ (not 70); block power = 11; $E\{3x_1\} = 495$; the shifted and reversed signals both have energy 55.
- **Program 4:** E_N rises quickly ($E_0 = 1$, $E_1 = 1.64$, $E_5 = 2.5869$, $E_{10} = 2.7573$) and then stays just below the red line at 2.7778; $E_{30} = 2.77775$.
- **Program 5:** the running power starts at 4, swings around 2 and settles at 2; the power over one period and over five periods are both 2.
- **Program 6:** P_N starts at $2/3$ ($N = 1$) and decreases towards $1/2$; $P = 0.52381$, 0.50249 and 0.50025 for $N = 10$, 100 and 1000 .

The command window shows (Programs 3 – 6):

```
E1 = 55, E2 = 15
E(x1+x2) = 100, E1+E2 = 70
Power of x1 over 5 samples = 11
E(3x1) = 495, 9*E1 = 495
E(x1(n-3)) = 55, E(x1(-n)) = 55
E_N(N=30) = 2.777775, 1/(1-a^2) = 2.777778
P one period = 2, P 5 periods = 2
Theoretical power A^2/2 = 2
Unit step, N = 10 : P = 0.52381
Unit step, N = 100 : P = 0.50249
```

Table 3.3 Sequences, energies and powers computed in the experiment

Quantity	Expected value	Observed (MATLAB)
Common index range	$-2 \leq n \leq 3$	
$x_1(n) + x_2(n)$	1, 2, 6, 3, 7, 1	
$x_1(n) \cdot x_2(n)$	0, 0, 9, -4, 10, 0	
E_1 (energy of x_1)	55	
E_2 (energy of x_2)	15	
$E\{x_1 + x_2\} / E_1 + E_2$	100 / 70	
Power of x_1 over $L_1 = 5$ samples	11	
$E\{3x_1(n)\}$	495	
$E\{x_1(n-3)\}$, $E\{x_1(-n)\}$	55, 55	
E_N of $0.8^n u(n)$, $N = 30$	2.777775	
Closed-form energy $1/(1-a^2)$	2.777778	
Power of $2\cos(2\pi n/16)$: 1 period / 5 periods	2 / 2	
Power of $u(n)$, $N = 10 / 100 / 1000$	0.52381 / 0.50249 / 0.50025	

Observations

Conclusion

In this experiment, two signals with different index ranges were lined up by zero-padding and then added and multiplied sample by sample. The energy of finite signals and of $0.8^n u(n)$ (2.777775, exact 2.777778) was found, the power of a sinusoid ($A^2/2 = 2$) and of the unit step ($1/2$) was calculated, and the signals were classified as energy or power signals. All results were verified using MATLAB.

Exercise:

1. For $x_1(n) = \{2, 1, -1, 3\}$ on $-1 \leq n \leq 2$ and $x_2(n) = \{1, 2, 1\}$ on $1 \leq n \leq 3$, find and plot $x_1(n) + x_2(n)$, $x_1(n) - x_2(n)$ and $x_1(n) \cdot x_2(n)$.
2. Find the energy of $x_1(n)$, $x_2(n)$ and $x_1(n) + x_2(n)$ from Exercise 1. Is $E\{x_1 + x_2\}$ equal to $E_1 + E_2$?

3. Repeat Program 4 for $a = 0.5$ and $a = 0.9$. Which one reaches its exact energy faster? Why?
4. Find the power of $x(n) = 3 \cos(2\pi n/10)$ over one period and compare it with $3^2/2$.
5. Find $P_N = (1/(2N + 1)) \sum x^2(n)$ over $-N \leq n \leq N$ for the ramp $x(n) = n u(n)$ with $N = 10, 20$ and 40 . Does it settle to a fixed value?
6. Generate $x(n) = \exp(j\pi n/8)$ for $0 \leq n \leq 15$ and show that its power is 1 (use `abs`).

Viva Questions:

1. Why can x_1 and x_2 of Program 1 not be added directly with `x1 + x2`?
2. What is zero-padding and why is it needed?
3. Why is the product $x_1(n) \cdot x_2(n)$ non-zero only for $0 \leq n \leq 2$?
4. What is the difference between `*` and `.*` in MATLAB?
5. Define the energy and the average power of a discrete-time signal.
6. Why is $E\{x_1 + x_2\} = 100$ and not $55 + 15 = 70$?
7. How does the energy change if a signal is multiplied by 3? Shifted? Reversed?
8. What is the power of $A \cos(\omega_0 n)$? Why is the power of $\cos(\pi n)$ equal to 1?
9. Why is the power of the unit step $1/2$?
10. The power of x_1 over its 5 samples is 11, but $x_1(n)$ is called an energy signal ($P = 0$). Why?
11. Can a non-zero signal be both an energy signal and a power signal? Why?
12. Is the ramp $r(n) = n u(n)$ an energy signal, a power signal or neither?

Connection to the Next Experiment:

Experiment 04 (Sampling, Quantization and Aliasing of Signals) uses the power idea from this lab. The quantization error $e(n)$ is treated as noise, and its *noise power* is $(1/L) \sum e^2(n)$ — the same block power as in topic 7. The quality of a quantizer is given by the *signal-to-quantization-noise ratio* $SQNR = 10 \log_{10}(P_{\text{signal}} / P_{\text{noise}})$ dB, where $P_{\text{signal}} = A^2/2$ for a sinusoid, as found here. Zero-padding and sample-by-sample multiplication are used again in Experiment 05 (convolution).